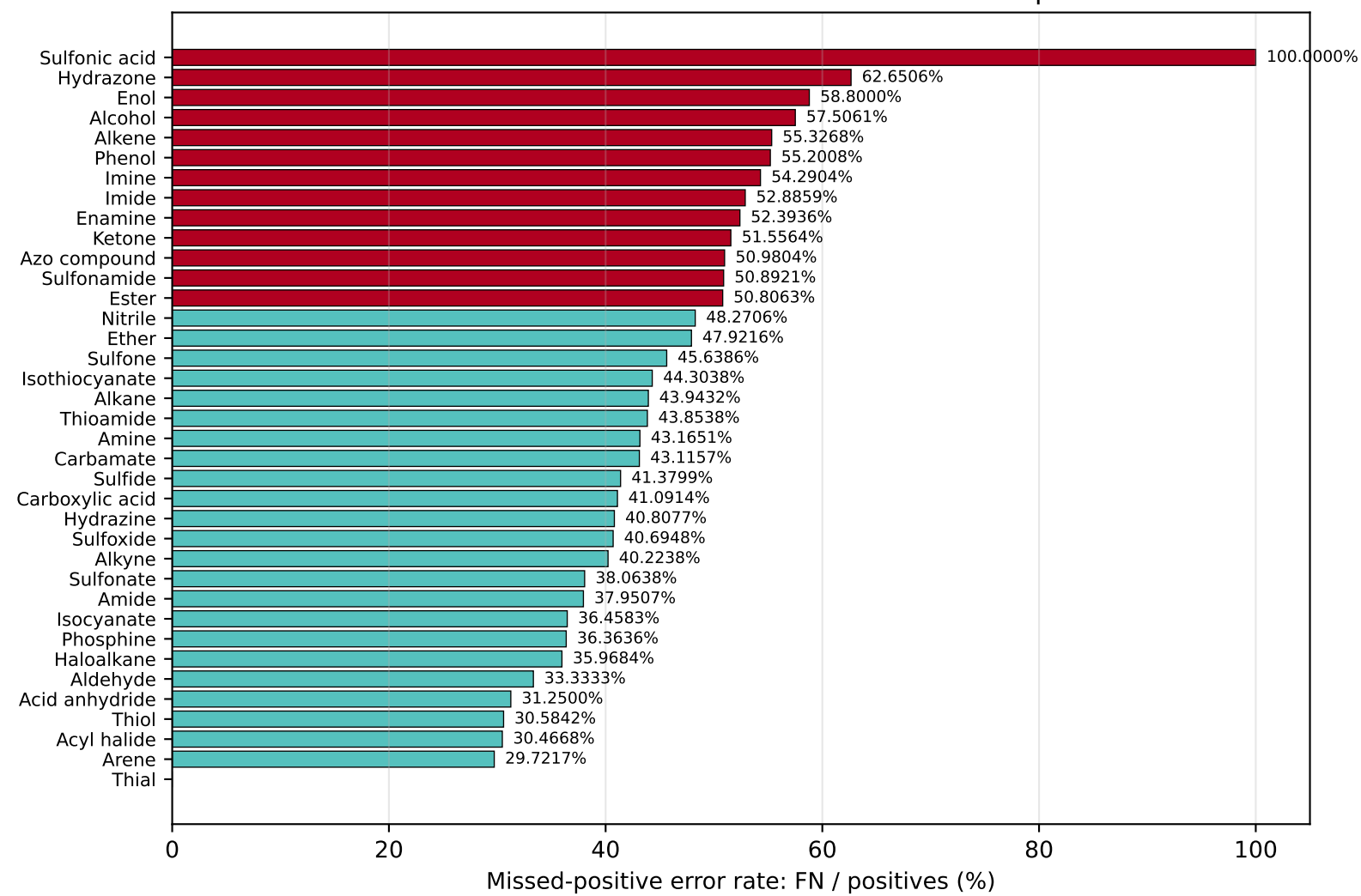
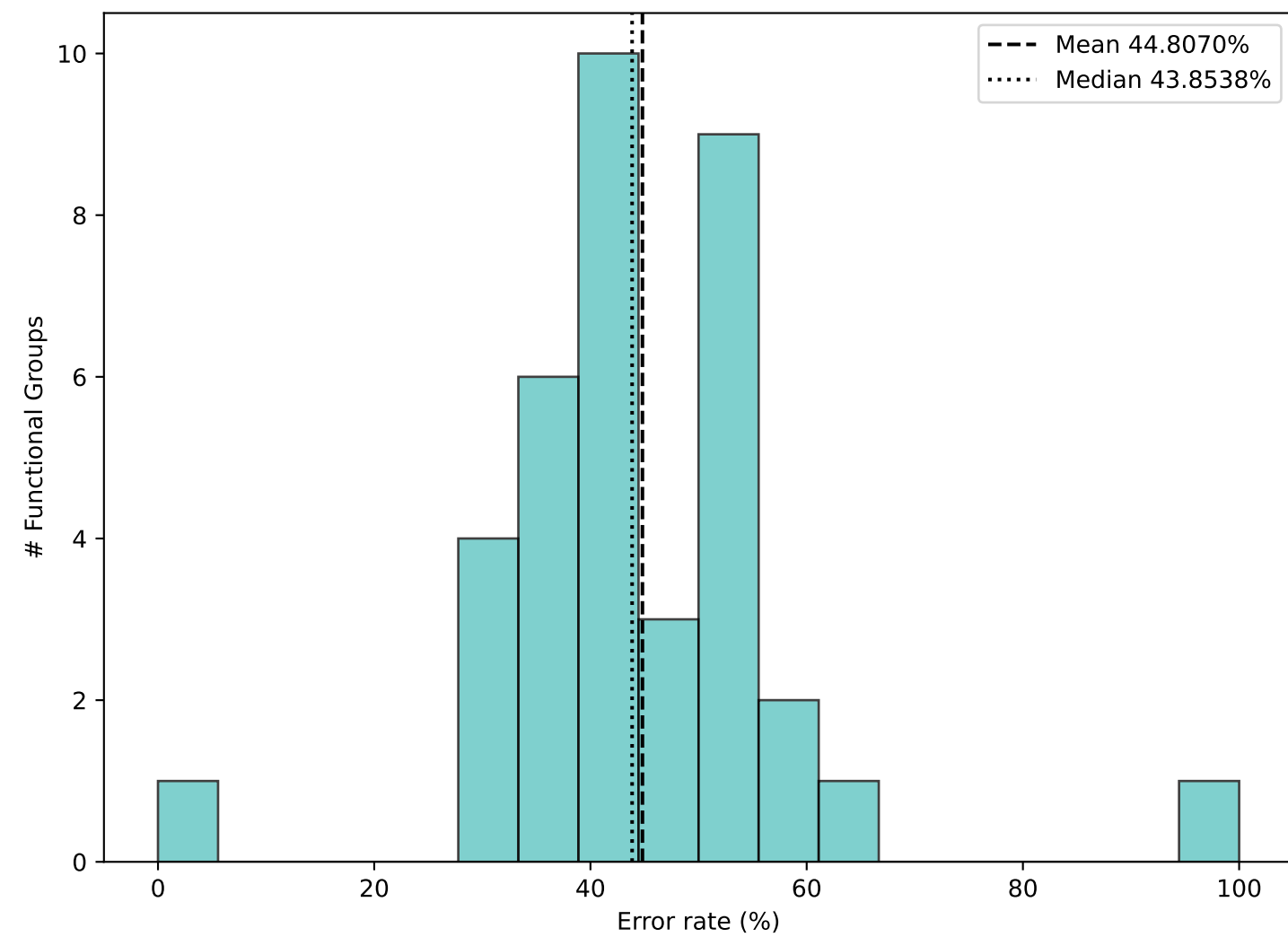


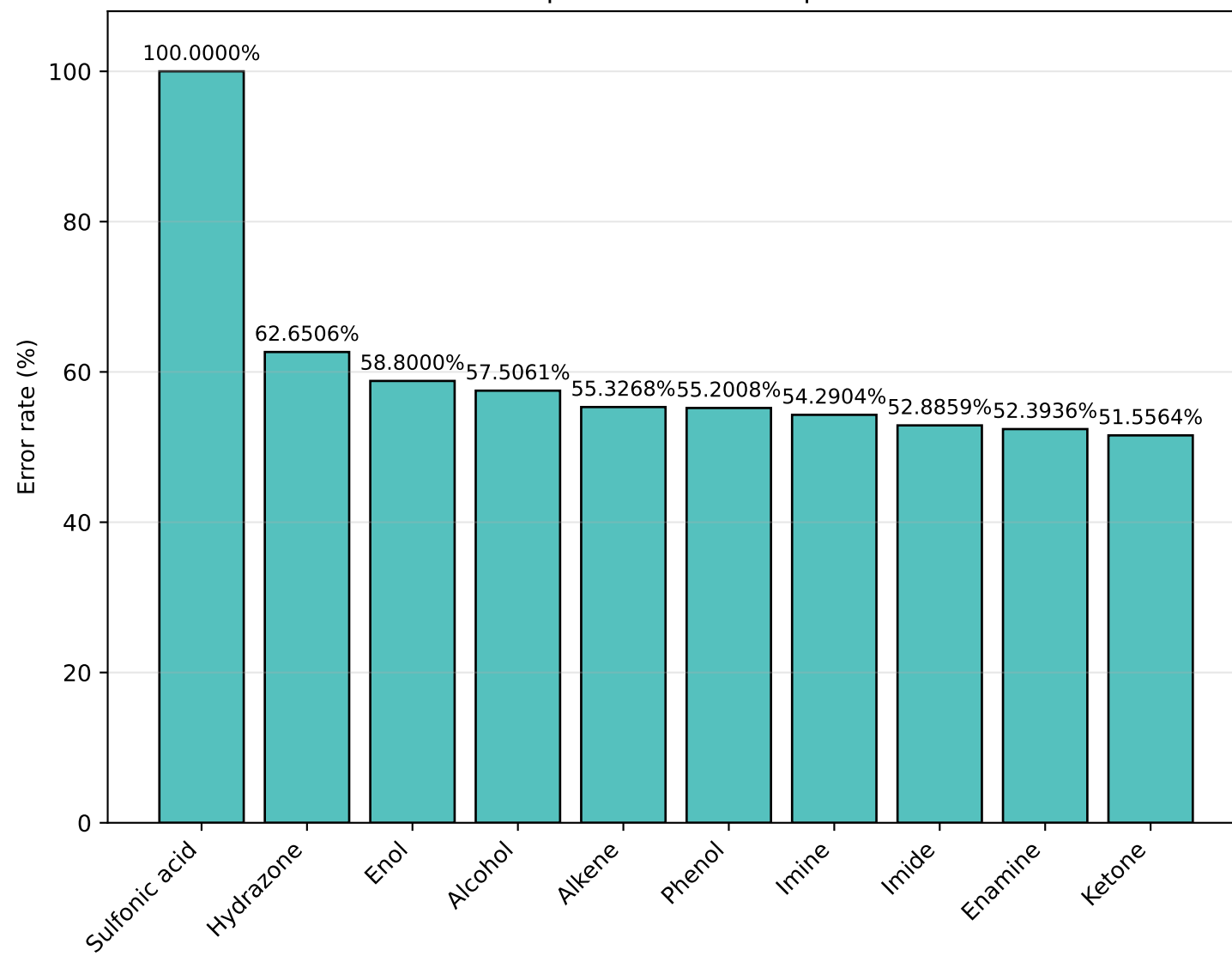
MPS MSMS+ Test Set - All 37 Functional Groups



Error-Rate Distribution



Top 10 Problem Groups



MPS MSMS+ TEST SET ERROR ANALYSIS

Global metrics:

F1 micro: 0.2971
F1 macro: 0.1745
Precision micro: 0.2003
Recall micro: 0.5751
Hamming accuracy: 0.6481

Error-rate statistics:

Mean: 44.8070%
Median: 43.8538%
Std: 14.6149%
Max: 100.0000% (Sulfonic acid)
Min: 0.0000% (Thial)

Problem classes:

Severe (>50%): 13 groups
High (20-50%): 23 groups
Moderate (5-20%): 0 groups
Low (<=5%): 1 groups